

IT3081 Statistical Modelling Practical Test Practice Bank

Statistical Modeling Support Group

Question 01: Importing, inspecting and cleaning data

The file `Hitters.csv` is supplied. Write R code to:

- (a) import the file as `hitters`;
- (b) display the first six observations, the structure and the dimensions;
- (c) count the missing values in each variable;
- (d) remove incomplete observations and store the result as `hitters_clean`;
- (e) confirm that no missing values remain and produce a numerical summary of `Salary`.

[10 marks]

Question 02: One-sample and independent two-sample t -tests

The following vectors record task-completion times in minutes.

```
method_A <- c(42, 39, 45, 41, 38, 44, 40, 43)
method_B <- c(47, 46, 43, 49, 45, 48, 44, 46)
```

Write R code to:

- (a) test whether the mean completion time for Method A differs from 40 minutes;
- (b) test whether the mean times of the two independent methods differ;
- (c) save both test results as objects and extract the test statistic, degrees of freedom and p -value;
- (d) write one decision statement for each test.

[10 marks]

Question 03: z -tests and population proportions

Use functions introduced in Lab Sheet 02.

- (a) For `scores <- c(102, 98, 105, 110, 96, 101, 107, 99)`, use `BSDA::z.test()` to test $H_0 : \mu = 100$, assuming the known population standard deviation is 6.
- (b) In a survey, 84 of 120 students support a new system. Use `prop.test()` to test whether the population support proportion is 0.60.
- (c) A second campus reports 63 supporters among 105 students. Test whether the two campus proportions differ.
- (d) For each analysis, extract or print the p -value and state the conclusion.

[12 marks]

Question 04: Variance, normality and correlation

Two numeric vectors `x` and `y` are already available in the R environment. Write code to:

- (a) draw a histogram and perform a Shapiro–Wilk test for each vector;
- (b) use an F -test to compare the two population variances;
- (c) produce a scatterplot of `y` against `x`;
- (d) calculate Pearson's correlation coefficient and conduct its significance test;
- (e) identify the two outputs needed to decide whether the linear association is statistically significant.

*[10 marks]***Question 05: One-way ANOVA and post-hoc comparison**

The following data compare scores under three teaching methods.

```
scores <- data.frame(  
  mark = c(62, 65, 68, 64, 66, 72, 75, 71, 74, 73,  
           78, 81, 79, 83, 80),  
  method = factor(rep(c("A", "B", "C"), each = 5))  
)
```

Write R code to:

- (a) display the group means and side-by-side boxplots;
- (b) apply Levene's test using `car::leveneTest()`;
- (c) fit the one-way ANOVA using `aov()` and display its table;
- (d) if the ANOVA is significant, apply Fisher's LSD procedure using `agricolae::LSD.test()`;
- (e) report which output indicates whether at least one population mean differs.

[12 marks]

Question 06: Simple and multiple linear regression

Load `MASS` and use the `Boston` data.

- (a) fit a simple regression predicting `medv` from `lstat`;
- (b) fit a multiple regression predicting `medv` from all remaining variables;
- (c) display both summaries and extract R^2 and adjusted R^2 ;
- (d) fit another model using all predictors except `chas`;
- (e) predict `medv` for a new observation whose predictor values are supplied in a correctly named data frame;
- (f) use adjusted R^2 , not ordinary R^2 alone, to compare the two multiple-regression models.

[14 marks]

Question 07: Train–test evaluation of a regression model

Using `MASS::Boston`, write R code to:

- (a) set the seed to 123 and randomly assign 80% of rows to training data;
- (b) fit `medv` on all other variables using only the training data;
- (c) predict the test responses;
- (d) calculate test MSE and test RMSE without using a specialised metric package;
- (e) calculate training RMSE in the same way;
- (f) state what pattern in the two RMSE values would suggest overfitting.

[14 marks]

Question 08: Factors, dummy variables and prediction

Use `mtcars`. Convert `am` into a factor with levels labelled `Automatic` and `Manual`. Then:

- (a) display the factor levels and the contrast matrix;
- (b) fit `mpg` as a function of `wt`, `hp` and `am`;
- (c) identify the reference transmission category from the R output;
- (d) obtain predictions for two otherwise identical cars, one automatic and one manual;
- (e) use the fitted coefficient to describe the adjusted difference between the two categories.

[12 marks]

Question 09: Checking multiple-regression assumptions

Fit `medv ~ .` using `MASS::Boston`. Write code to perform all of the following:

- (a) plot residuals against fitted values and add a horizontal reference line at zero;
- (b) draw a normal Q–Q plot of the residuals and apply the Shapiro–Wilk test;
- (c) use `lmtest::dwtest()` to check residual independence;
- (d) use `car::vif()` to assess multicollinearity;
- (e) state the residual-plot pattern expected under homoscedasticity;
- (f) identify which output would warn you about highly correlated predictors.

[14 marks]

Question 10: Best-subset selection

Use the cleaned ISLR::Hitters data to predict Salary.

- (a) remove incomplete observations;
- (b) run best-subset selection using `leaps::regsubsets()` with a suitable `nvmax`;
- (c) save the model summary and plot or display C_p , BIC and adjusted R^2 against model size;
- (d) identify the model size preferred by minimum C_p , minimum BIC and maximum adjusted R^2 ;
- (e) print the coefficients of the model chosen by BIC.

[14 marks]

Question 11: Forward and backward stepwise selection

Again use the cleaned Hitters data.

- (a) fit one forward-selection model and one backward-selection model using `regsubsets()`;
- (b) select the preferred size for each method using BIC;
- (c) extract the corresponding coefficient sets;
- (d) compare the variables retained by the two procedures;
- (e) write one sentence explaining why the selected sets need not be identical.

[12 marks]

Question 12: Ridge regression with cross-validation

The objects `x_train`, `x_test`, `y_train` and `y_test` have been created from the supplied car data. Complete the code and answer the tasks.

```
library(glmnet)
ridge_cv <- cv.glmnet(_____, _____, alpha = ___)
best_lambda <- ridge_cv$_____
ridge_pred <- predict(ridge_cv, s = best_lambda,
                     newx = _____)
```

- (a) fill all blanks so the code performs cross-validated ridge regression;
- (b) plot the cross-validation curve;
- (c) calculate test RMSE;
- (d) extract the coefficients at the selected value of λ ;
- (e) compare the ridge test RMSE with the test RMSE from an ordinary multiple-regression model fitted to the same split.

[14 marks]

Question 13: LASSO estimation and variable selection

Use the same training and test matrices as in Question 12.

- (a) fit a LASSO coefficient path using `glmnet()` and plot it against $\log(\lambda)$;
- (b) use `cv.glmnet()` to select `lambda.min`;
- (c) predict the test responses and calculate RMSE;
- (d) extract the coefficients at `lambda.min`;
- (e) list the selected predictors with non-zero coefficients and the dropped predictors with zero coefficients;
- (f) state the coding change that distinguishes LASSO from ridge in `glmnet()`.

[14 marks]

Question 14: Binary logistic regression, odds ratios and classification

Use `mtcars`, where `am` is coded 0 for automatic and 1 for manual. Write code to:

- (a) fit a binary logistic model `am ~ wt + hp`;
- (b) display the model summary and confidence intervals;
- (c) calculate odds ratios by exponentiating the coefficients;
- (d) obtain fitted probabilities and convert them to predicted classes using a 0.50 cut-off;
- (e) construct a confusion matrix and calculate classification accuracy;
- (f) interpret the sign of the coefficient of `wt`.

[14 marks]

Question 15: Comparing null and full logistic models

For the supplied leukaemia-remission data, `REMISS` is binary and the predictors include `LI` and `TEMP`.

- (a) fit the null model containing only an intercept;
- (b) fit the model `REMISS ~ LI` and then the model `REMISS ~ LI + TEMP`;
- (c) display coefficient estimates, confidence intervals and odds ratios for the full model;
- (d) compare the null and full models using an appropriate likelihood-based comparison in R;
- (e) use `ResourceSelection::hoslem.test()` with observed outcomes and fitted probabilities;
- (f) state what a large Hosmer–Lemeshow p -value suggests.

[14 marks]

Question 16: LDA and QDA on the same train–test split

Use `MASS::lda()`, `MASS::qda()` and the `iris` data.

- (a) set the seed to 2222 and create an 80:20 train–test split;
- (b) fit LDA using all four measurements to predict `Species`;
- (c) fit QDA using the identical training and test observations;
- (d) predict the test classes for both models;
- (e) create a confusion matrix and calculate accuracy for each model;
- (f) select the better model for this split and justify the choice using the computed results.

[14 marks]

Question 17: Naive Bayes classification

Using the same `iris` split from Question 16:

- (a) fit a Naive Bayes classifier using `e1071::naiveBayes()`;
- (b) predict the test classes;
- (c) construct the confusion matrix and calculate accuracy;
- (d) predict a new flower with measurements (5.9, 3.0, 5.1, 1.8), in the data-set column order;
- (e) compare its test accuracy with the LDA and QDA results obtained from the same split.

[12 marks]

Question 18: Multinomial logistic regression

Use `nnet::multinom()` with `iris`.

- (a) use the same seed and 80:20 split as in Question 16;
- (b) fit a multinomial logistic model predicting `Species` from all measurements;
- (c) display the model coefficients and identify the reference response category used by R;
- (d) predict the test classes, form the confusion matrix and calculate accuracy;
- (e) predict the class of the new flower given in Question 17;
- (f) explain why binary logistic regression is not sufficient for this response.

[14 marks]

Question 19: Ordinal logistic regression

Use `MASS::housing`, where `Sat` is ordered and `Freq` contains frequency weights.

- (a) confirm that `Sat` is an ordered factor, correcting it if necessary;
- (b) fit an ordinal logistic model using `MASS::polr()` with `Inf1`, `Type` and `Cont` as predictors, `weights = Freq`, and `Hess = TRUE`;
- (c) display the model summary and confidence intervals;
- (d) identify the regression coefficients separately from the threshold parameters;
- (e) obtain predicted satisfaction classes;
- (f) state why the frequency weights and `Hess = TRUE` are used.

[14 marks]

Question 20: GLM model choice and interpretation

Complete all three short coding tasks using the datasets named below.

(i) Poisson model: Using `warpbreaks`, fit `breaks ~ wool + tension` with a log-linked Poisson model. Display the summary, obtain incidence-rate ratios and predict the mean count for supplied predictor levels.

(ii) Exponential curve: Using `Puromycin`, fit

$$\log(\text{rate}) = \beta_0 + \beta_1 \text{conc.}$$

Recover $a = \exp(\hat{\beta}_0)$ and $b = \hat{\beta}_1$, write the fitted equation $\hat{y} = a \exp(bx)$, and produce fitted values.

(iii) Gamma model: Using `USArrests`, fit `Assault ~ Murder + UrbanPop + Rape` with a log-linked Gamma model. Exponentiate one slope and explain its multiplicative effect.

For each task, add one sentence explaining why the selected response distribution or transformation is appropriate for the response variable.

[18 marks]